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Contributions to the Knowledge of Hydrogen Peroxide. On the Action of Hydrogen Peroxide on Hexamethylenetetramine.

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Depending on whether hydrogen peroxide is allowed to act on the free base hexamethylenetetramine in aqueous solution, or on salts of the base in concentrated solutions, one obtains entirely different reaction products. Hexamethylenetetramine in concentrated aqueous hydrogen peroxide solution gives a new salt of the latter, a **hexamethylenetetramine–hydrogen peroxide** of the formula $(CH_2)_6N_4 \cdot H_2O_2$. From hexamethylenetetramine salts, hydrogen peroxide produces the **hexamethylene triperoxide diamine** — already obtained by another route by Legler[1] and later by Baeyer and Villiger[2] —

$$\begin{array}{c} CH_2 \cdot O \cdot O \cdot CH_2 \\ N\,CH_2 \cdot O \cdot O \cdot CH_2\,N \\ CH_2 \cdot O \cdot O \cdot CH_2 \end{array}$$

(the two nitrogen atoms joined by three $-CH_2 \cdot O \cdot O \cdot CH_2-$ bridges).

This differing behavior of hexamethylenetetramine lies in the nature of hydrogen peroxide. The acidic reaction of hydrogen peroxide has been pointed out from many sides. Sodium peroxide, barium peroxide, and so on are therefore often described as salts of hydrogen peroxide. In contrast to this view, Carrara and Bringhenti[3] pointed out that under certain conditions hydrogen peroxide behaves like a monobasic acid. From the decomposition voltages of sulfuric-acid-containing hydrogen peroxide solutions they concluded the presence of H^+ and $O \cdot OH^-$ ions.

In fact, a series of reactions is known in which hydrogen peroxide behaves like a monobasic acid. In aqueous solution, in the dissociated state, hydrogen peroxide behaves like a **hydrogen peracid** capable of forming salts with one molecule of a base, in which one hydrogen atom can be replaced by alkyl or other organic groups. In this respect it behaves like the first member of a homologous series, much as formic acid only apparently occupies an exceptional position within the series of monobasic fatty acids. From the hydrogen peracid are derived, for example: $CH_3 \cdot O \cdot O \cdot H$ methyl hydroperoxide, $C_2H_5 \cdot O \cdot O \cdot H$ ethyl hydroperoxide, and so on, as well as performic acid, peracetic acid, etc. Salts of this acid include, for example, the sodyl hydroxide $Na \cdot O \cdot O \cdot H$ described by Tafel[4], and the carbamide–hydrogen peroxide $CO(NH_2)_2 \cdot H_2O_2$ first prepared by Tanatar[5], which has recently come onto the market under the name “Hyperol.” That these salts do not involve crystal–hydrogen peroxide (i.e. peroxide of crystallization) is evident from their behavior on warming in vacuum: with rising temperature, decomposition occurs rather than the hydrogen peroxide distilling over undecomposed, as is the case with organic salts containing crystal–hydrogen peroxide. The aforementioned hexamethylenetetramine–hydrogen peroxide also belongs to this class of salts of the hydrogen peracid. On evaporation in vacuum, even with varying proportions of the components, the compound containing equal molecules always separates first from the aqueous solution. A replacement of the two hydrogen atoms by bases or metals in aqueous solution has never been observed.

As is evident from the large number of known peroxide compounds, hydrogen peroxide by no means always behaves like a monobasic acid. If the dissociation of the acid is suppressed by a large excess of salt or also by acids to a minimum, the formation of **peroxides** proceeds — that is, of compounds in which both hydrogen atoms are replaced by new atoms or atomic groups.

Thus, with exclusion of water, from sulfuric anhydride (2 mol) and hydrogen peroxide (1 mol) persulfuric acid is formed: $OH{\cdot}SO_2{\cdot}O{\cdot}O{\cdot}SO_2{\cdot}OH$, whereas in the presence of water monopersulfuric acid (Caro's acid) is formed: $OH{\cdot}SO_2{\cdot}O{\cdot}OH$. This too is to be regarded as a derivative of the monobasic hydrogen peracid.

Similarly, the conditions now also apply to hexamethylenetetramine, which is known to form highly concentrated, supersaturated solutions with a series of organic acids. The dissociating action of water on the hydrogen peroxide is suppressed, and with elimination of ammonia and substitution of the two hydrogen atoms the already mentioned hexamethylene triperoxide diamine is formed. The substance is insoluble in water; therefore, once formed, no subsequent dissociation with water occurs.

The yield depends on the amount of undissociated hydrogen peroxide. The reaction therefore does not occur at all in dilute solutions, despite the insolubility of the substance in water. Since the acid does not primarily take part in the reaction, it is immaterial whether one uses acetic acid, oxalic acid, citric acid, or other acids. In each case the named peroxide is formed in more or less good yield, provided that as little water as possible is used as solvent. The best yields were obtained with **citric acid**.

Legler obtained the compound from the "lamp acid" $OH{\cdot}CH_2{\cdot}O{\cdot}O{\cdot}CH_2{\cdot}OH$ (diformal peroxide hydrate), formed during the slow oxidation of ethyl ether, by reaction with ammonia. Here, then, the finished peroxide was already present in the starting material. Baeyer and Villiger obtained considerably better yields by allowing formaldehyde and hydrogen peroxide to react in a concentrated solution of ammonium sulfate (50 g ammonium sulfate dissolved in 50 g of 3% hydrogen peroxide). In all three cases the principle of formation is the same: either a concentrated salt solution must be present, or the substituted hydrogen peroxide (in the case of the lamp acid, formed with exclusion of water by oxidation of ether) must enter the reaction already formed.

Experimental

Hexamethylenetetramine–Hydrogen Peroxide, $(CH_2)_6N_4{\cdot}H_2O_2$.

The ca. 30% hydrogen peroxide used in the following experiments was obtained from the ordinary commercial 3% product by evaporation on a vigorously boiling water bath with addition of a minimal amount of an acid. By means of a strong air current, the accelerated evaporation ensured that the temperature did not rise above 60–70°. On enrichment up to 30%, work was carried out with a yield averaging 90–95% of theory. The concentration can readily be raised in good yield up to 60% and 70% hydrogen peroxide; but the heat liberated in the reaction acts harmfully at higher concentrations, which is why work was mostly carried out with 30% hydrogen peroxide.

To prepare the salt, one molecule of hexamethylenetetramine is dissolved in a slight excess of 30% hydrogen peroxide and, after complete dissolution, concentrated in vacuum. To avoid decomposition, the vacuum should be taken at least strong enough that the boiling point comes to lie no higher than 40–50°. The salt separates as coarse, colorless crystalline masses, which can be separated from the mother liquor by suction and dried completely over phosphorus pentoxide.

The yield becomes nearly quantitative if the solution is evaporated completely to dryness. Addition of very small amounts of acid during evaporation increases the stability of the salt. Larger amounts (over 3%) are to be avoided, since otherwise syrupy liquids easily form that solidify only with difficulty and often undergo spontaneous decomposition after some time, which propagates through the entire mass. Similar phenomena also occur when the temperature rises above 60° during concentration. In this case ammonia, trimethylamine, ammonium formate, etc. are formed in large amounts.

The salt is readily soluble in water and alcohol, as well as in most organic solvents. Concentrated sulfuric acid decomposes the salt with vigorous deflagration; from concentrated hydrochloric acid, chlorine is evolved by oxidation. In general, the solution of the salt shows the characteristic properties of the components.

0.9938 g substance: 45.42 cc ½-*n.* NH_3. — 0.5392 g substance: 61.13 cc $^1/_{10}$-*n.* $KMnO_4$. — 0.2188 g substance: 25.10 cc $^1/_{10}$-*n.* $KMnO_4$.

Calc. N 32.25, H_2O_2 19.54.

Found N 32.03; H_2O_2 19.59, 19.58.

The hydrogen peroxide content can be determined directly with permanganate in acidic solution, since the oxidation of the formaldehyde proceeds only very slowly at ordinary temperature. The first drop of permanganate that colors the solution pink for about three to four seconds is to be taken as the endpoint of the hydrogen peroxide titration. Titrations with potassium iodide and titration of the liberated iodine with thiosulfate give the same values.

In nitrogen determination by Kjeldahl, the substance must first be dissolved in a little water and then the sulfuric acid added gradually, since with concentrated acid the reaction proceeds far too violently.

Hexamethylene Triperoxide Diamine, $(CH_2)_6N_2O_6$.

Since inorganic acids mostly act decomposingly on the peroxide, they are not suitable for its preparation. The best yields were obtained with **citric acid**.

If 28 g hexamethylenetetramine and 42 g citric acid are dissolved in 140 g of 30% hydrogen peroxide, the solution warms somewhat and gradually deposits abundant amounts of hexamethylene triperoxide diamine as small, regular, rhombic crystals. Washed with water and then with alcohol, one obtains 27.5 g, corresponding to a yield of 66% of theory, referred to hexamethylenetetramine.

The properties of this interesting substance have already been described in detail by Legler (l.c.) and by Baeyer and Villiger (l.c.), so I would only refer to those publications. Besides by its properties, the peroxide was identified by analysis:

0.1991 g substance: 0.2521 g CO_2, 0.0914 g H_2O. — 0.3005 g substance: 28.72 cc $n/_{10}$-NH_3. — 0.2948 g substance: 28.12 cc $n/_{10}$-NH_3. — 0.1313 g substance: 38.40 cc $n/_{10}$-$Na_2S_2O_3$.

Calc. C 34.60, H 5.81, N 13.46, act. O 23.07, O 23.06 (diff.).

Found C 34.53, H 5.14, N 13.39, 13.36; O 23.39, 23.59.

Like so many aliphatic peroxides, hexamethylene triperoxide diamine is a highly **explosive** substance, although its preparation is entirely without danger. Nevertheless it is not advisable to store the substance dry. It is best covered over with water, by which it is not further attacked.

Through the kindness of Privy Councillor Will-Neubabelsberg I was afforded a series of investigations of this substance as an **explosive**[6], on which I will report briefly in order to warn against the enormous danger of careless handling.

In friction tests in an unglazed porcelain mortar, strong localized explosions occurred. Touched with a burning match, deflagration ensues, sometimes with violent detonation. With a burning fuse in a test tube, the substance detonates with shattering of the glass. In loose soil a small blasting crater was thrown up. When 0.05 g of substance in a test tube was heated in a Wood's metal bath at a rate of temperature increase of about 20° per minute, a

violent explosion with shattering of the glass occurred at 139°. 0.5 g, thrown into a glowing iron dish, deflagrates violently.

In tests of explosion sensitivity with the drop hammer, a drop height of one centimeter at a hammer weight of 500 g was usually sufficient to bring the substance to explosion.

10 g brought to explosion in a Trauzl lead block with a No. 8 mercury fulminate cap gave a net cavity expansion of 242 cc. The expansion would presumably have been considerably greater had the cavity of the lead block not been almost completely filled owing to the small volume of the substance, so that the covering sand layer was only very thin.

Most interesting were the experiments carried out to investigate its initiating action. In a No. 8 copper shell, 0.5 g of substance was — under suitable precautions, of course — subjected to a slowly rising pressure in order to produce an initiator (detonator). At a pressure of about 250 atmospheres the substance exploded, and the shell cleanly pierced through a bronze base 3.5 mm thick, a cloth lying underneath in several layers, and the roughly 1 cm thick wooden floor of the apparatus.

From these experiments it thus emerges that this is a most dangerous substance which, for example, possesses considerably more dangerous and stronger explosive properties than **mercury fulminate** (cavity expansion in the lead block 77 cc).

In experimenting with these substances, the utmost caution is therefore required.

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[6] The investigations were carried out by Dr. Burkard in the laboratories of the Central Office for Scientific Investigation, G.m.b.H., Neubabelsberg. I should like to express here my most obliged thanks to Privy Councillor Will and Dr. Burkard for their kind support.

[1] L. Legler, *Ber.* **18**, 3343 [1885]. [2] Baeyer and Villiger, *Ber.* **33**, 2479 [1900]. [3] Carrara and Bringhenti, *G.* **33**, 362 [1903]. [4] Tafel, *Ber.* **27**, 816 & 2297 [1894]. [5] Tanatar, *Ž.* **40**, 376 [1908].

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